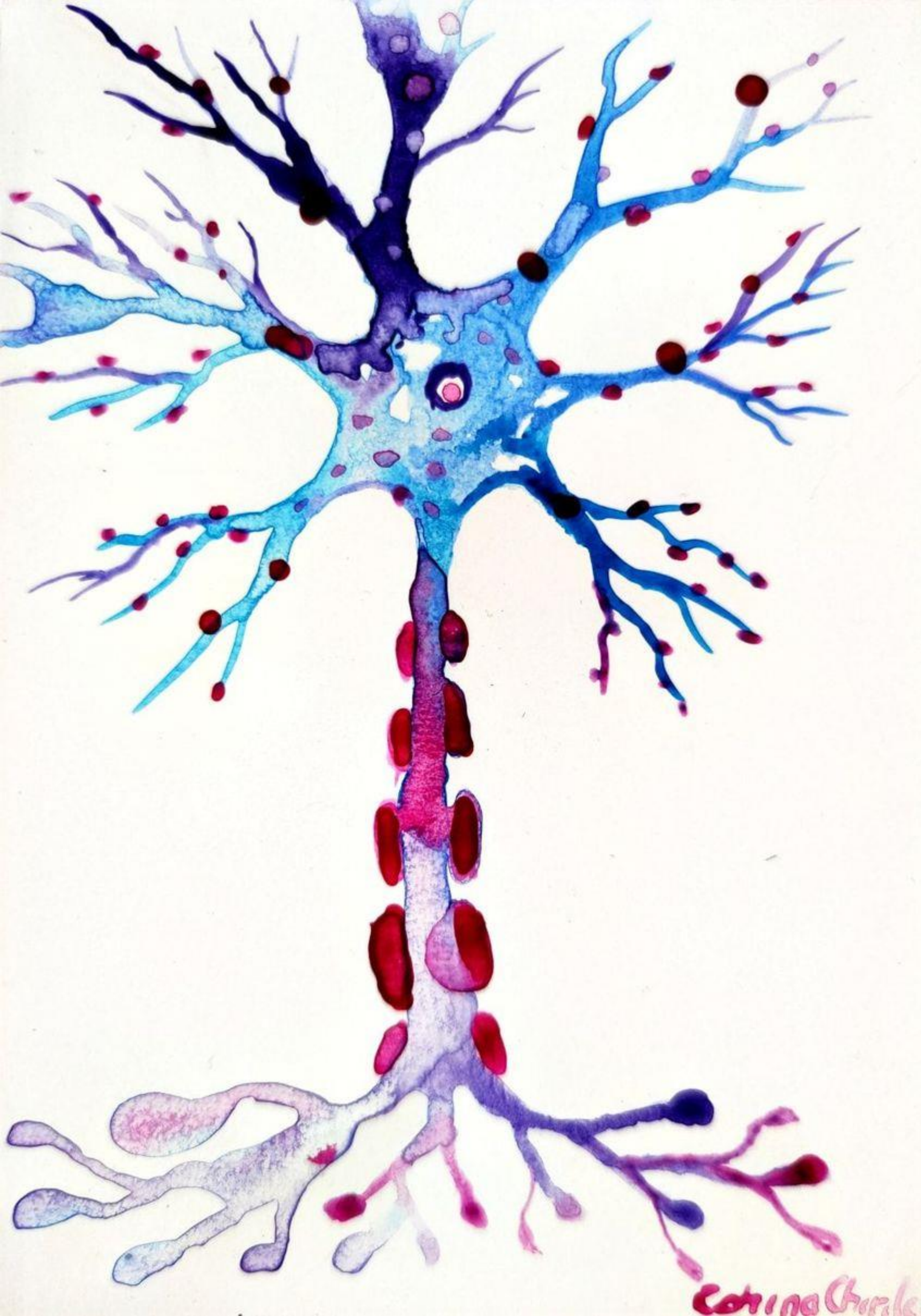
Neuroscience & Psychology Department Presentation

March 28, 2024



Investigating changes in tripartite synapses within the spinal cord of a novel ALS mouse model

Emma Butler

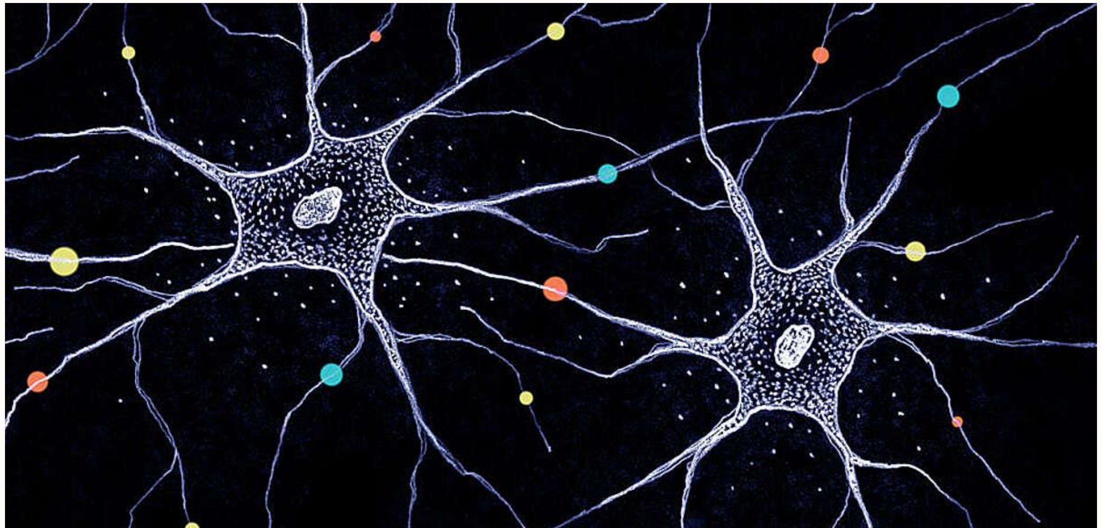
Quick review

- What is a neuron?

Quick review

- What is a neuron?

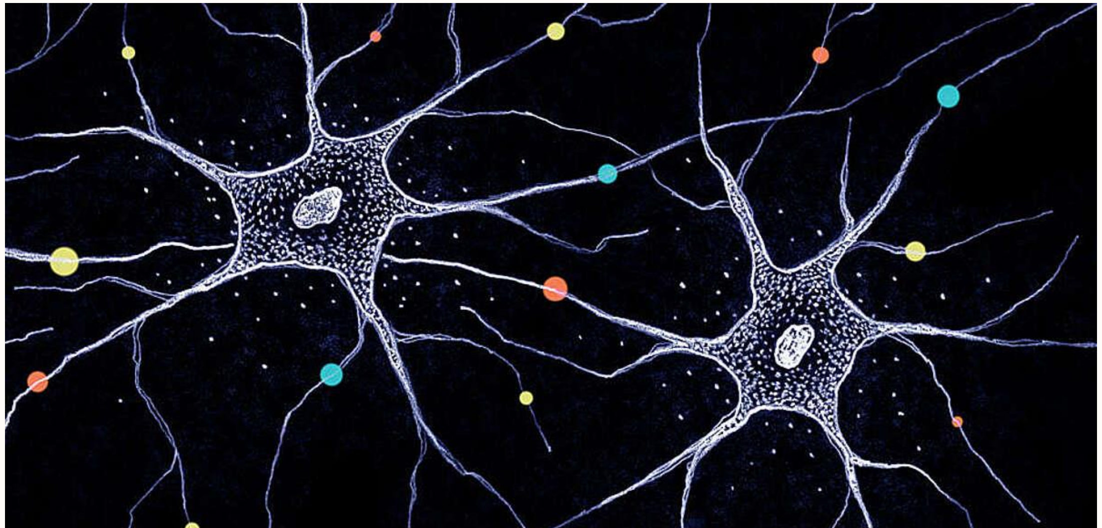
- Cells within the nervous system that send and receive information through electrochemical impulses



Quick review

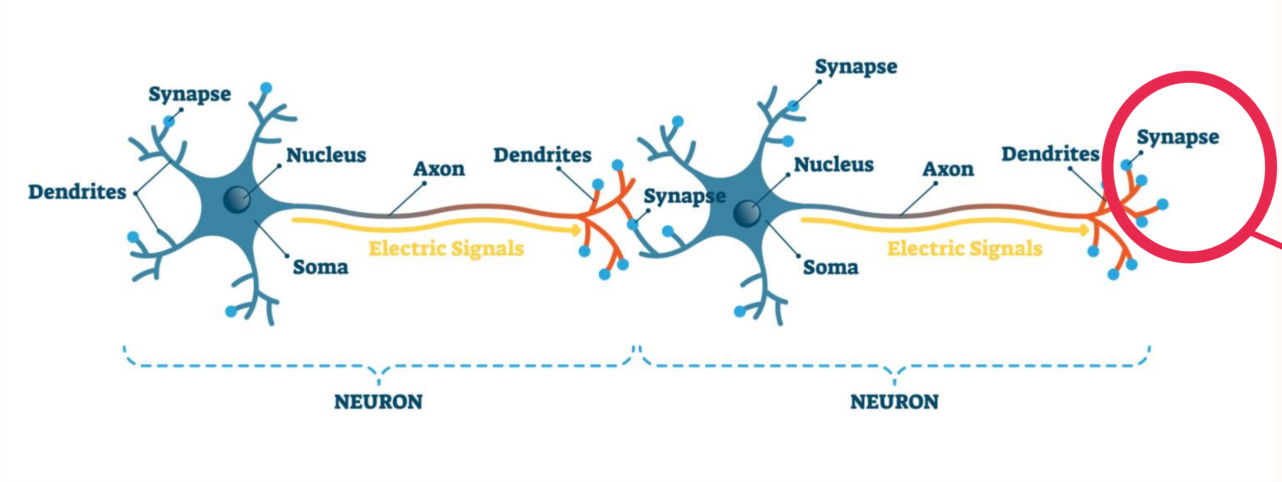
- What is a neuron?
- What is a synapse?

- Cells within the nervous system that send and receive information through electrochemical impulses

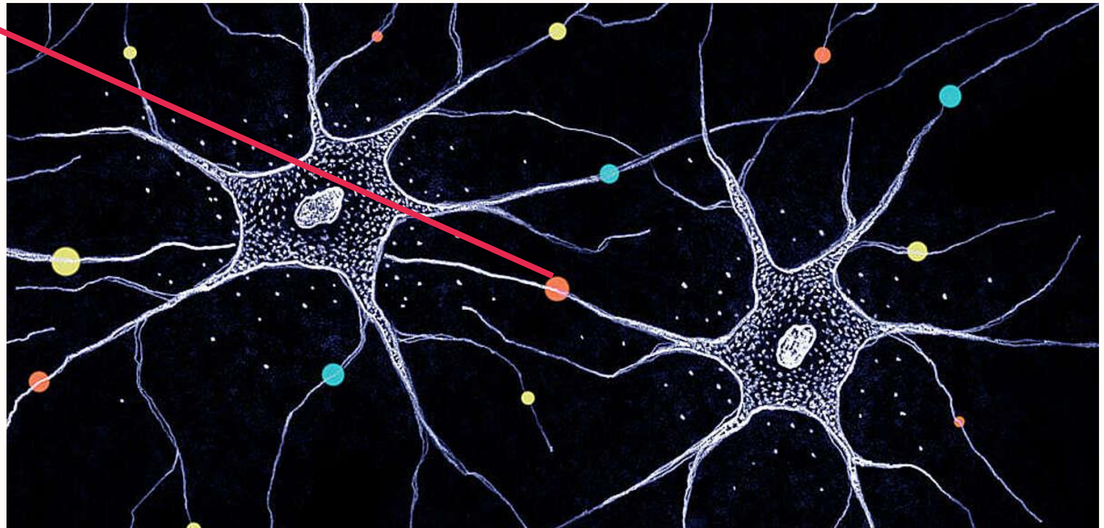


Quick review

- What is a neuron?
- What is a synapse?

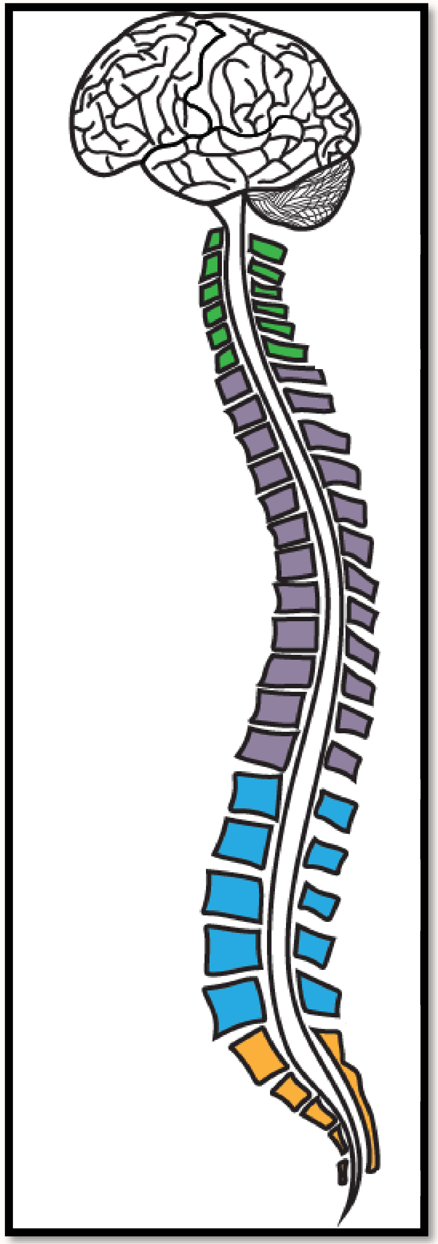


- Cells within the nervous system that send and receive information through electrochemical impulses
- Diverse structures that facilitates chemical communication from one neuron to the next

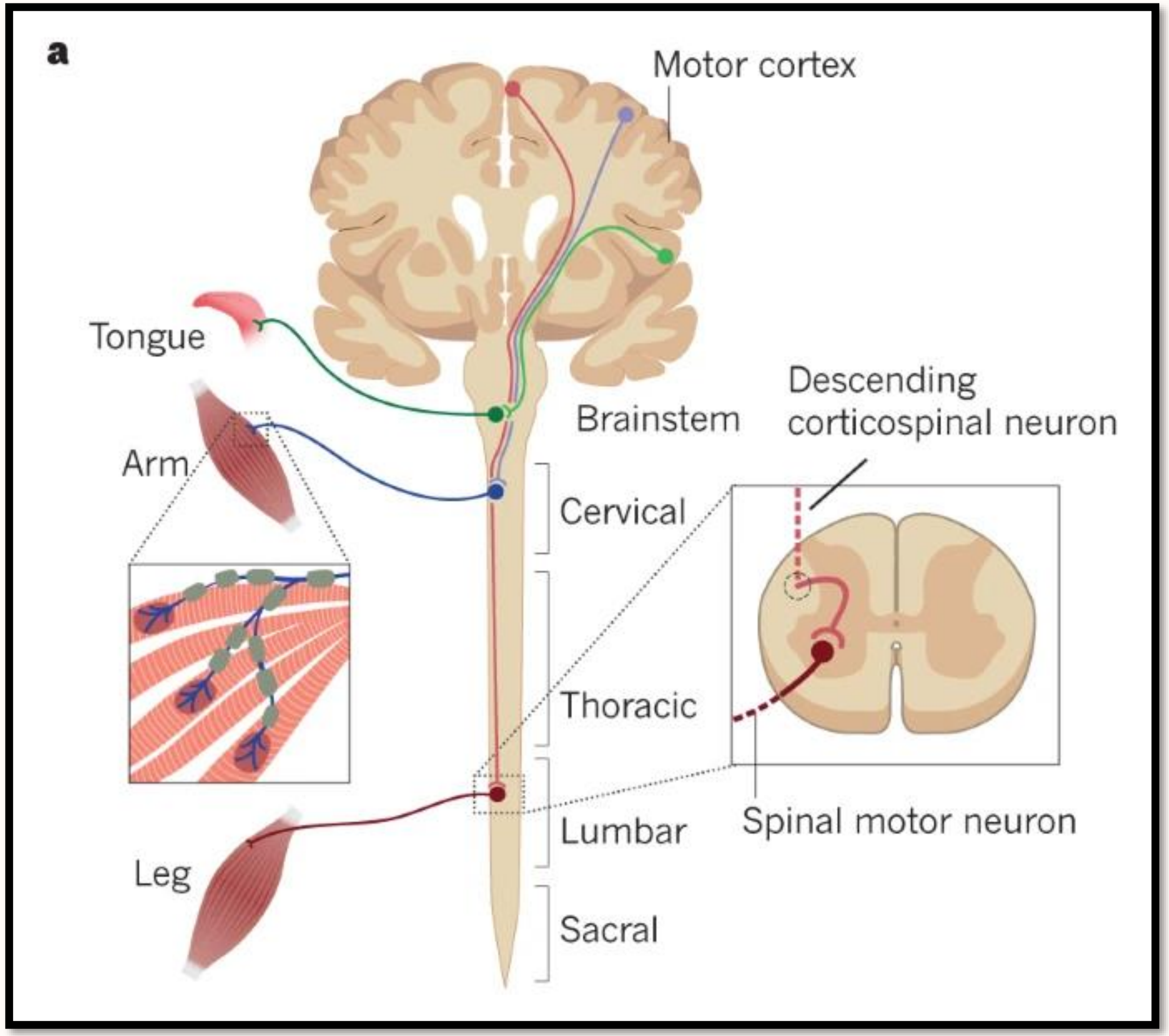


Amyotrophic Lateral Sclerosis (ALS)

- Progressive neurodegenerative disease
- Highly complex
- No cure
- Typically affects older adults and the elderly (40-60 years old)[1]
- More than 200,000 people around the world are living with ALS[1]
- Loss of motor neurons in the motor cortex, brain stem, and spinal cord, leading to weakness that progresses to muscle atrophy, paralysis, and eventual death[2,3,4]
- Approximately 80% of patients die within three to five years of diagnosis[2,3,4]

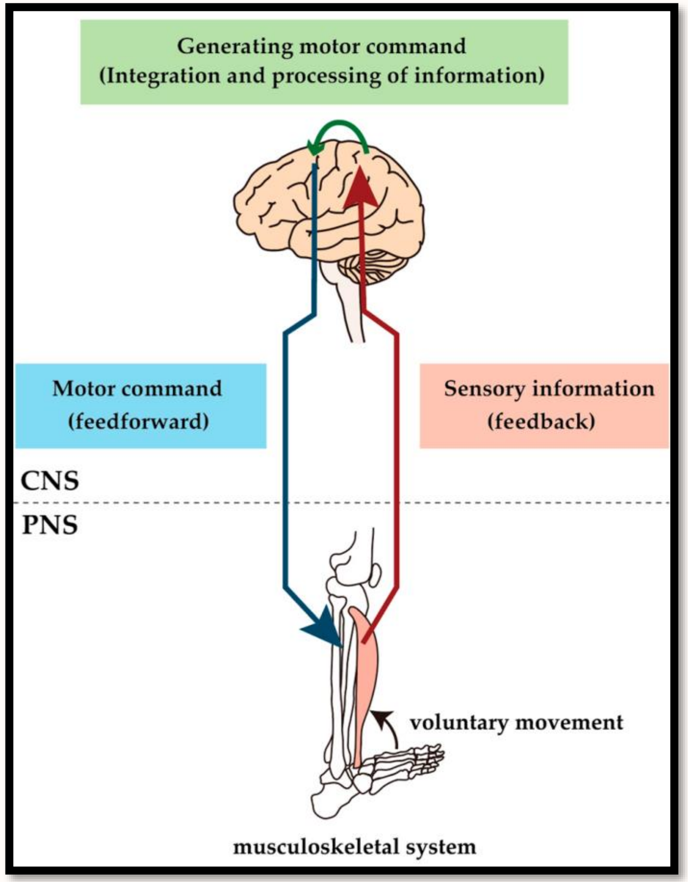


Parts of the nervous system affected by ALS[5]



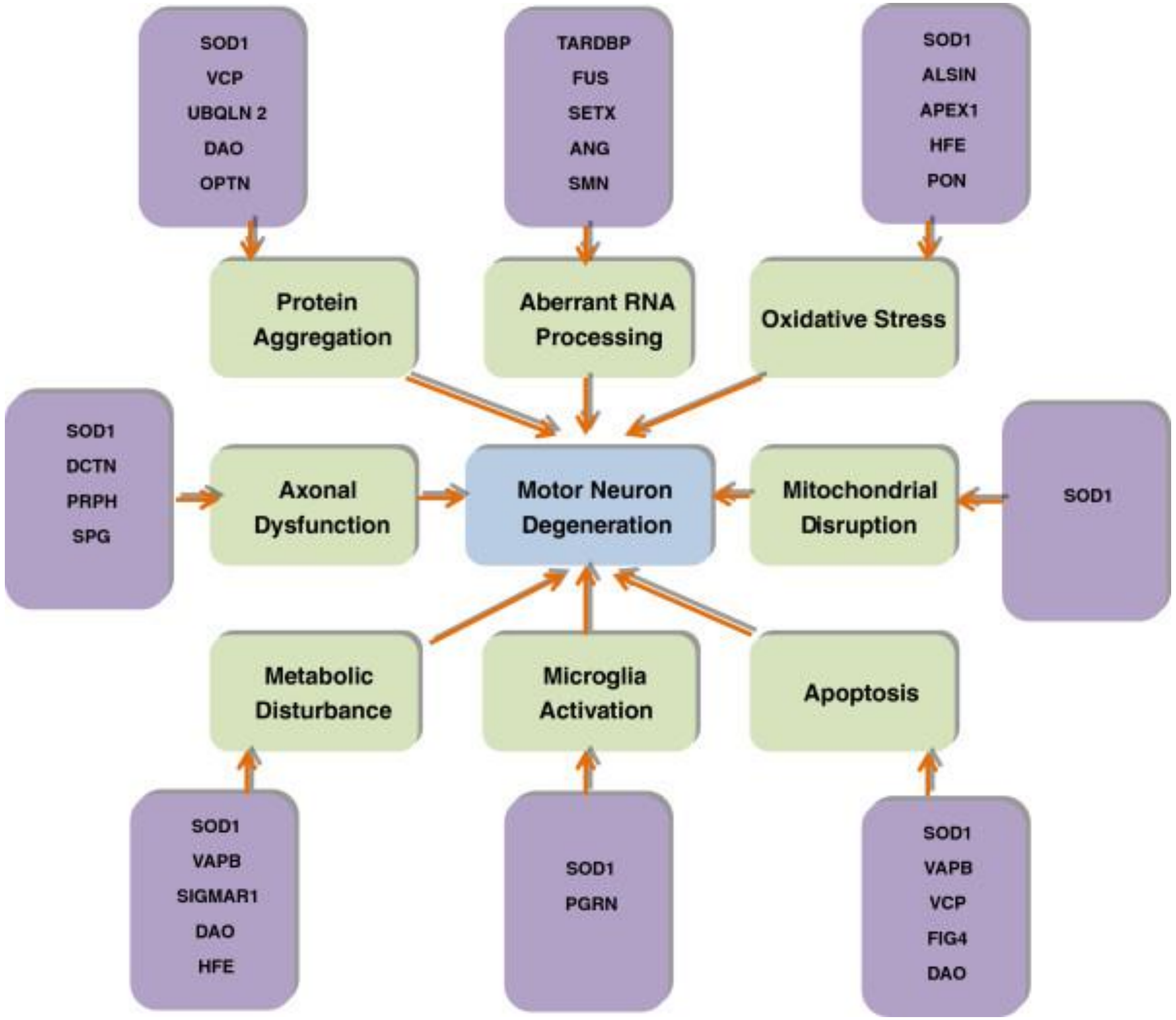
Amyotrophic Lateral Sclerosis (ALS)

- ALS alters the connection between the nervous system and muscles
- Riluzole and edavarone are the only two FDA-approved drugs for ALS treatment
  - They extend patient survival by less than a year[5,6]



Causes of ALS

- 10% of cases are familial (fALS) and 90% are sporadic (sALS)[7,8]
- Scientists have narrowed down at least **40** genes associated with sALS and fALS
  - > 20 genes associated with fALS alone[9]
- These genetic mutations cause complex molecular cascades along various pathways that eventually disrupt the normative functioning and interactions between neuronal and non-neuronal regulatory cells[9]
  - Motor neuron degeneration



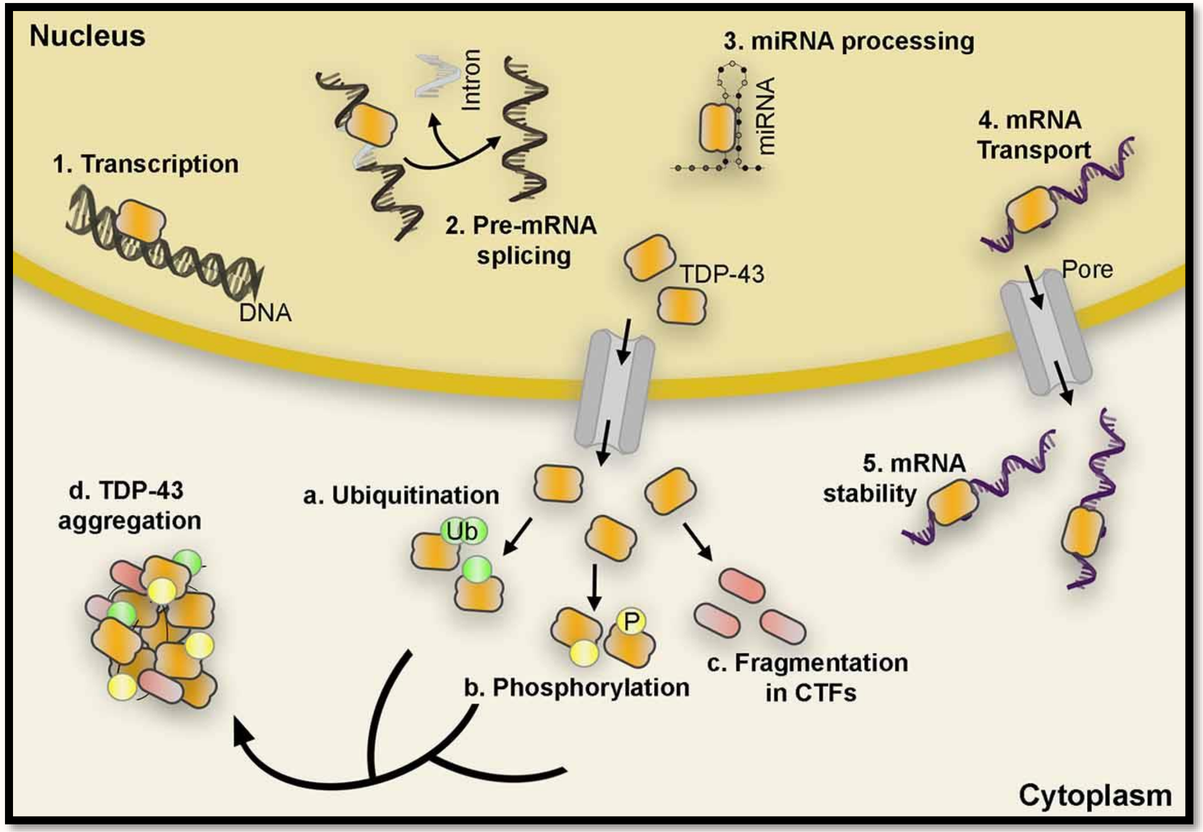
Causes of ALS

- 4-5% caused by TDP-43 mutation
- Neurological TDP-43 pathology seen in 95-97% of sALS and fALS cases[10,11]

Gene	Protein	Onset	Inheritance	Clinical feature	Other diseases caused by the gene
SOD1	Cu/Zn SOD-1	Adult	AD/AR	Typical ALS	NA
Alsin	Alsin	Juv	AR	Slowly progressive, predominantly UMN signs like limb, & facial spasticity	PLS IAHSP
Unknown	Unknown	Adu	AD	Typical ALS with limb onset especially lower limb	NA
SETX	Senataxin	Juv	AD	Slowly progressive, distal hereditary motor neuropathy with pyramidal signs	SCAR 1 and AOA2
SPG 11	Spatacsin	Juv	AR	Slowly progressive	HSP
FUS	Fused in Sarcoma	Juv/Adu	AD/AR	Typical ALS	NA
VAPB	VAPB	Adu	AD	Typical and atypical ALS	SMA
ANG	Angiogenin	Adu	AD	Typical ALS, FTD and Parkinsonism	NA
TARDBP	DNA-binding protein	Adu	AD	Typical ALS	NA
FIG 4	Phosphoinositide-5phosphatease	Adu	AD	Rapid progressive with prominent corticospinal tract signs	CMT 4 J
OPTN	Optineurin	Adu	AD/AR	Slowly progressive with limb onset and predominant UMN signs	Primary Open Angle Glaucoma
VCP	VCP	Adu	AD	Adult onset, with or without FTD	IBMPFD
UBQLN2	Ubiquilin 2	Adu/Juv	XD	UMN signs proceeding LMN signs	NA
SIGMAR1	SIGMAR1	Juv	AR	Juvenile onset typical ALS	FTD
unknown	unknown	Adu	AD	ALS with FTD	FTD
C9ORF72	C9ORF72	Adu	AD	ALS with FTD	FTD
DCTN1	Dynactin	Adu	AD	Distal hereditary motor neuropathy with vocal paresis	NA
Other rare-occurring ALS genes					
Unknown	Unknown	Adu	AD	Typical ALS with limb onset especially lower limb	NA
Unknown	Unknown	Adu	AD/AR	Typical ALS	NA
DAO	DAO	Adu	AD	Typical ALS	NA

TDP-43

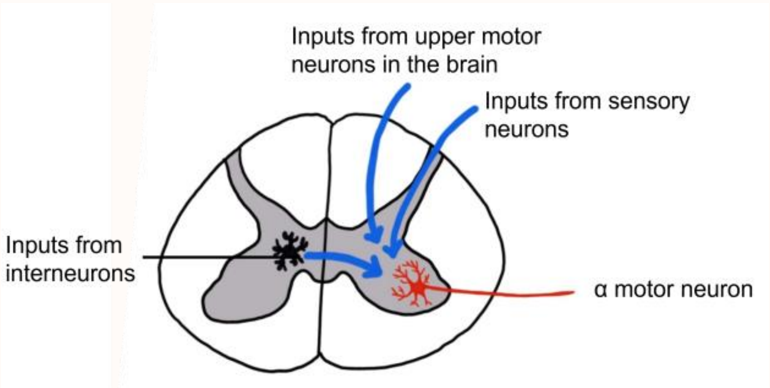
- TAR DNA-binding protein 43
- Helps regulate RNA metabolism[12,13,14]
  - Transcription
  - Splicing
  - mRNA transport
  - mRNA stability



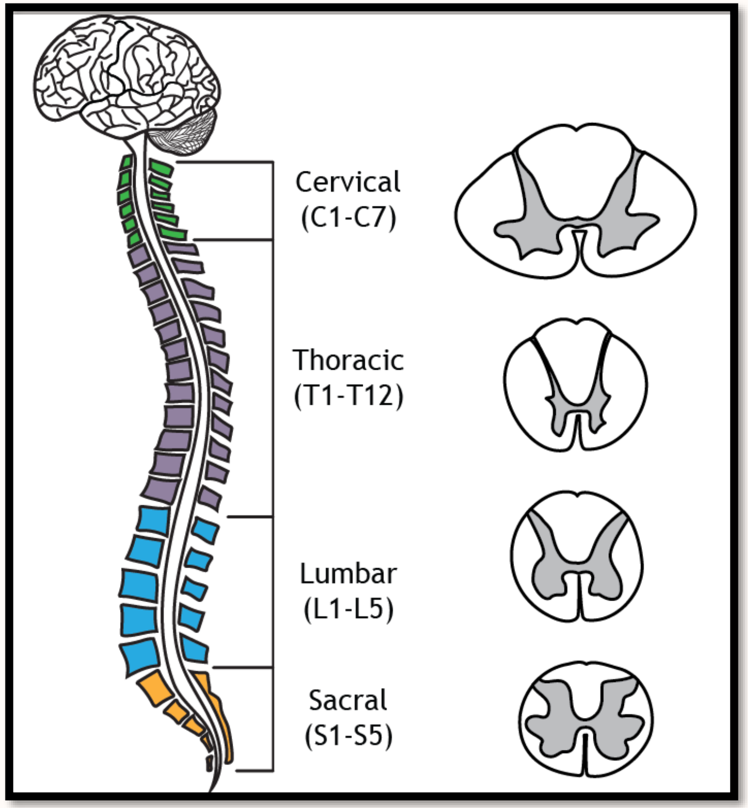
- Hyperphosphorylation, misfolding and abnormal localisation can lead to TDP-43 aggregation outside the cell nucleus[14,15]

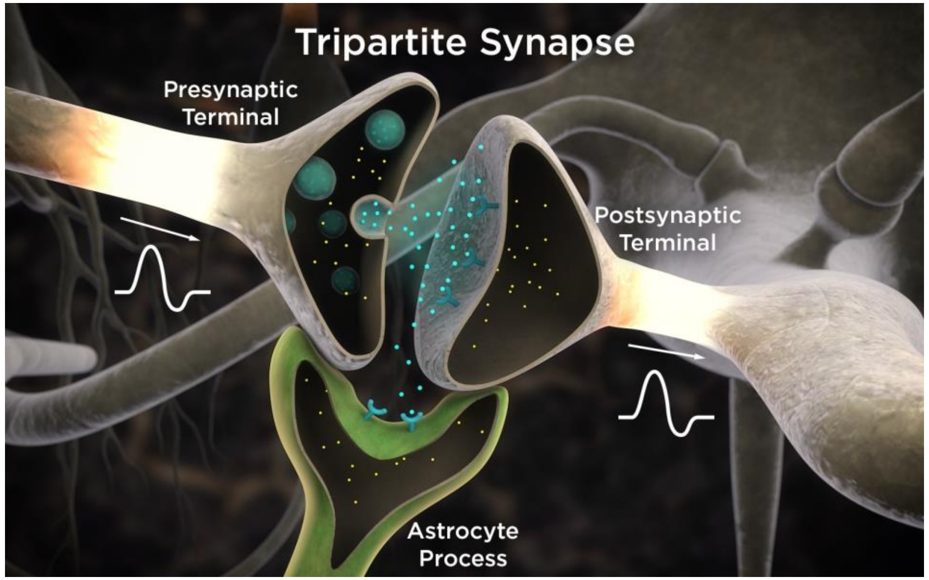
Potential mechanism

Synapses



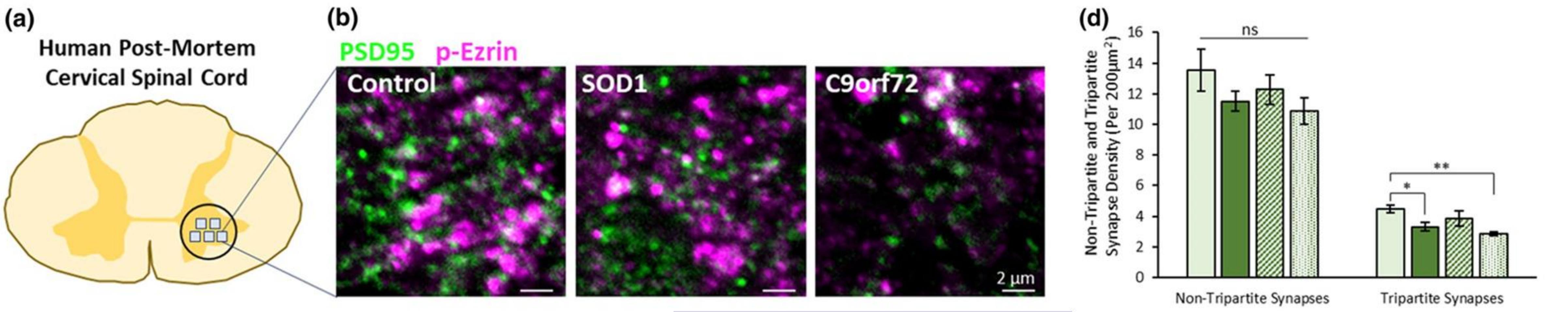
- In the early disease stages, synapses are changing[4]
- Substantial loss of synapses in lumbar and cervical spinal cord before the loss of motor neurons[16]
- Synaptic Diversity
  - Excitatory synapses[19,20]
  - Cholinergic synapses[18]
  - Inhibitory Synapses[19]





Tripartite Synapses

- Specialised chemical synapses contacted by astrocytes
- Astrocytes are glial cells (non-neuronal)
- Non-cell-autonomous disease mechanisms
- Selective vulnerability found in ALS SOD1 mouse model and human postmortem tissue[24,25]



Recap

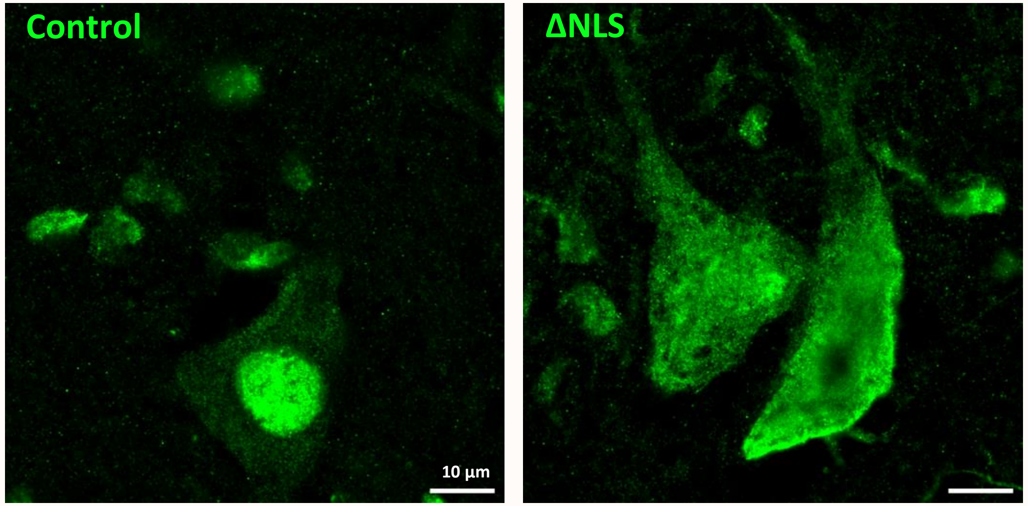
- ALS affects motor neurons in the brain and spinal cord
- Most ALS cases are sporadic and exhibit TDP-43 pathology
- Synaptic changes and loss are seen in the early disease stages
- Excitatory synapses that are contacted directly by astrocytes (tripartite synapse) show selective vulnerability in ALS

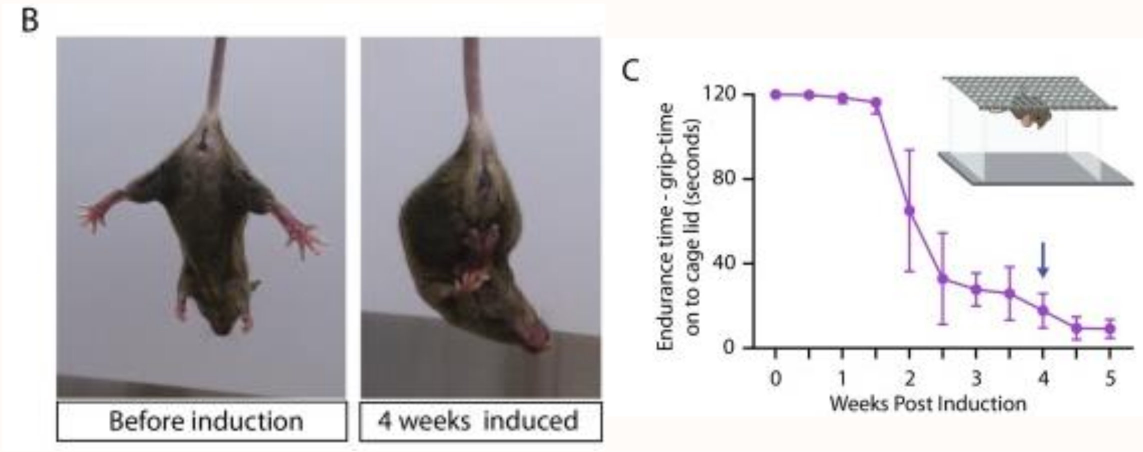
My Research Project

- Find out if excitatory tripartite synapse loss is conserved in a novel ALS mouse model
- Examine if other tripartite synapse types are also affected by ALS

TDP43ΔNLS Mouse Model

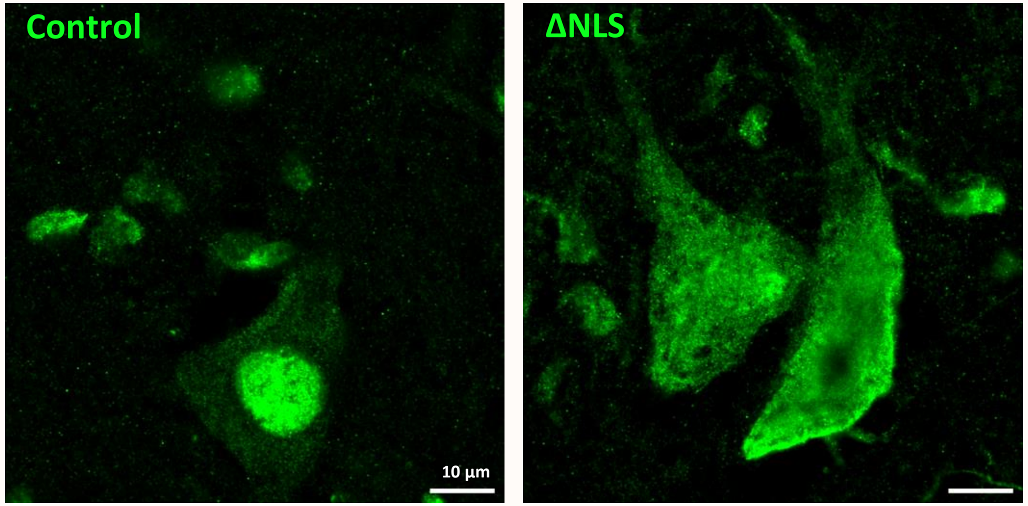
- TDP-43 devoid of nuclear localisation signal (NLS)
- Bigenic line
    - hTDP-43 x NEFH-tTA line 8[26,27]
- Controlled by doxycycline[28]
    - cytoplasmic aggregation, cell atrophy, and impaired motor function
- Claire Meehan's Lab - University of Copenhagen
- Not entirely clear why or how this aggregation leads to disease[29]

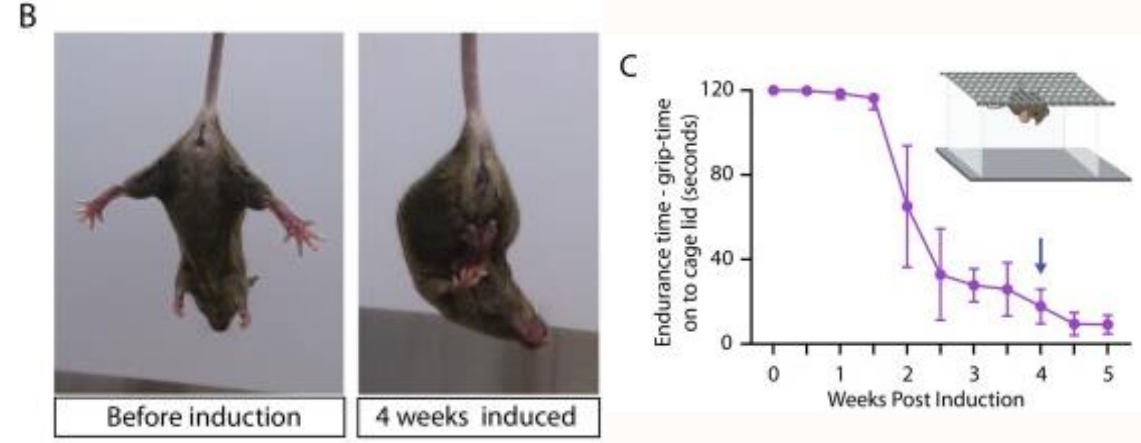




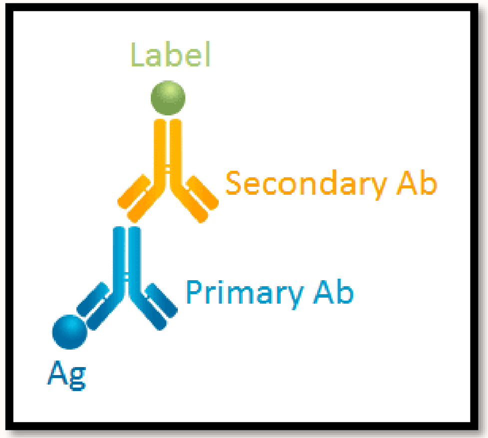
TDP43ΔNLS Mouse Model

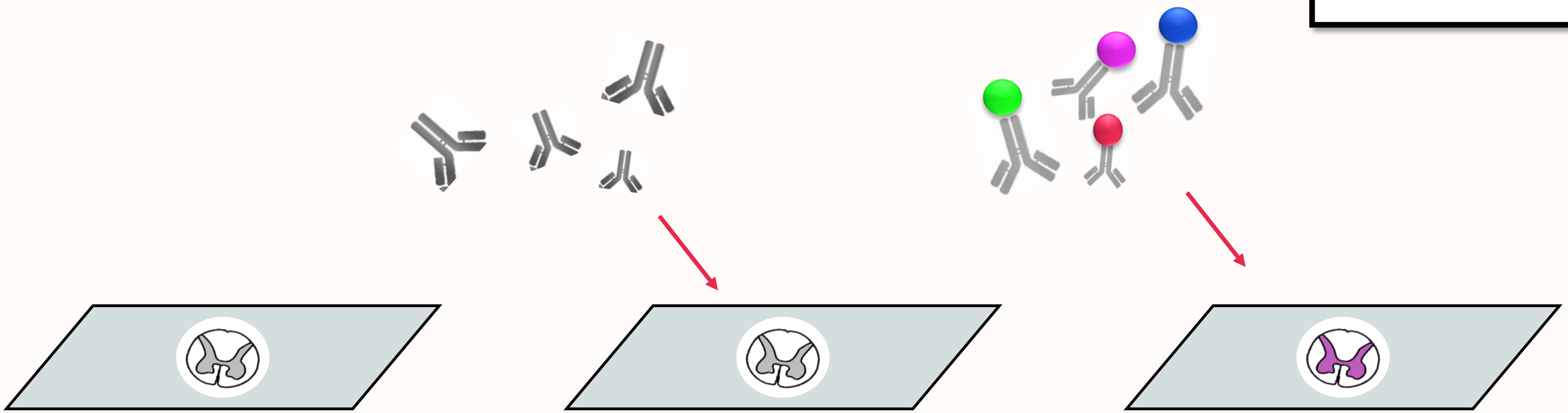
- ΔNLS induction at 8 weeks (N = 12)
  - Bigenic mice (n = 6) experienced ALS symptomology – reduced grip strength, limb clasping upon suspension, muscle atrophy
  - Transgenic controls (n = 6) did not show ALS motor phenotypes[30]
- Mice were euthanised after 4 weeks
- L2-3 lumbar segments were cryosectioned, adhered to Leica Xtra Adhesive histology slides, and stored at -80°C





Immunohistochemistry (IHC)





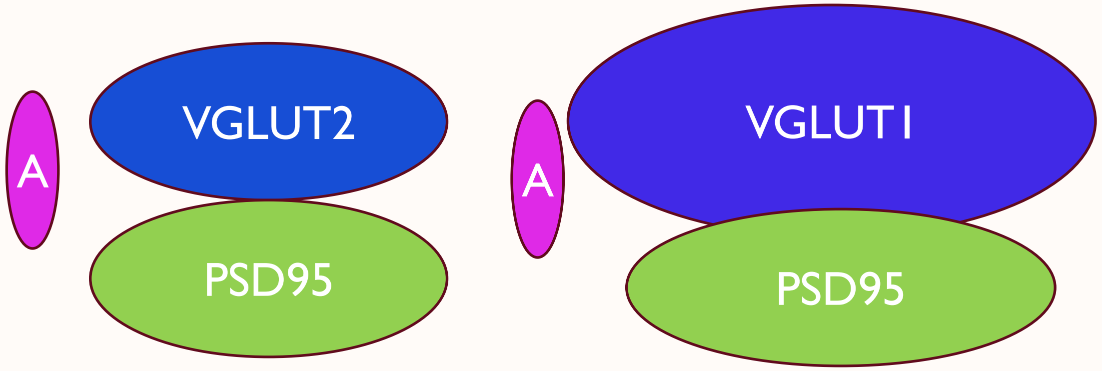
1) Incubate 37°C
2) Wash in 1% PBS
3) Apply "block/perm" solution

4) Apply primary antibodies
5) Store for 48 hrs at 4°C

6) Wash in 1% PBS
7) Add secondary antibodies
8) Apply coverslips

Immunohistochemistry (IHC)

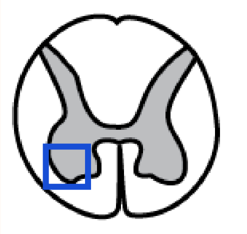
- PSD-95
  - Post-synaptic density scaffold protein
  - Tethers glutamate receptors and various signaling molecules[31]
- VGlut1
  - Vesicular glutamate transporter 1
  - Large excitatory boutons
  - 20% of glutamate transporters in spinal cord
  - Associated with mechanoreceptor terminals of dorsal afferents[32]
- VGlut2
  - Vesicular glutamate transporter 2
  - Small excitatory boutons
  - 80% of glutamate transporters in spinal cord[32,33]
  - Local spinal neuron terminals
- p-Ezrin
  - Phosphorylated Ezrin
  - Astrocytic terminals

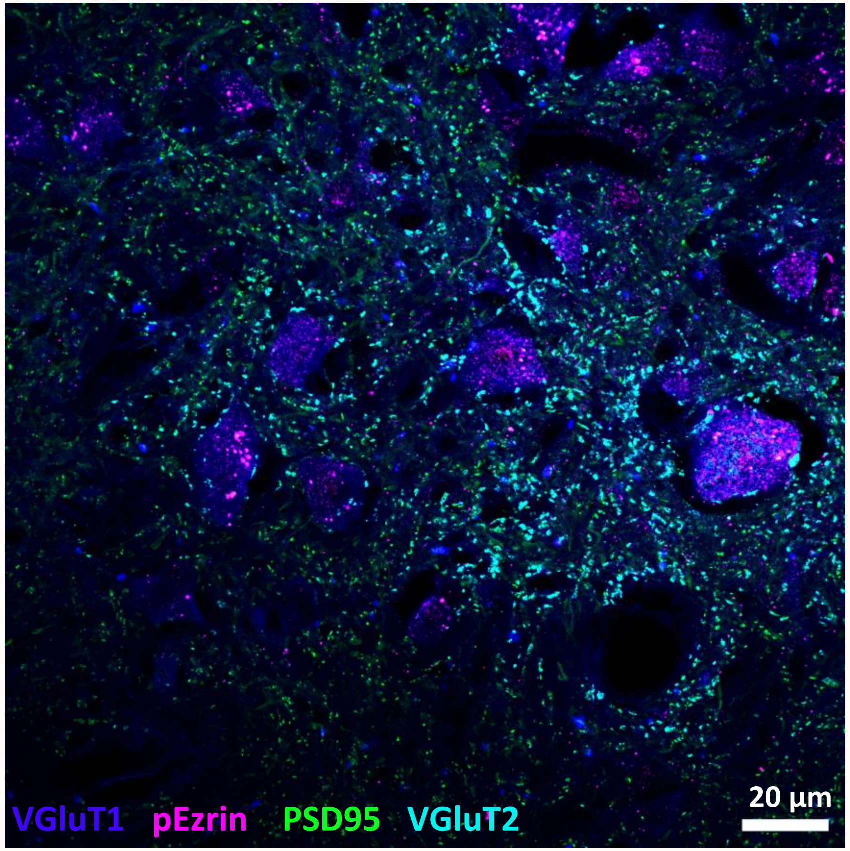


Primary Antibody Target	Species	Company	Code	Secondary Antibody/Emission Wavelength (nm)
PSD95	Camel nanobody	Synaptic Systems	N3702-At488-L	488
VGlut1	Guinea Pig	Millipore	ab5905	Dk anti-GP Alexa Fluor 647
Vglut2	Mouse	Abcam	ab79157	Dk anti-Ms Alexa Fluor 405+
p-Ezrin	Rabbit	Abcam	ab47293	Dk anti-Rb AlexaFluor 555

Imaging and Analysis

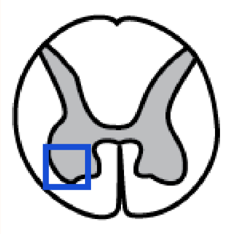
- Zeiss LSM800 Airyscan
  - High-resolution imaging

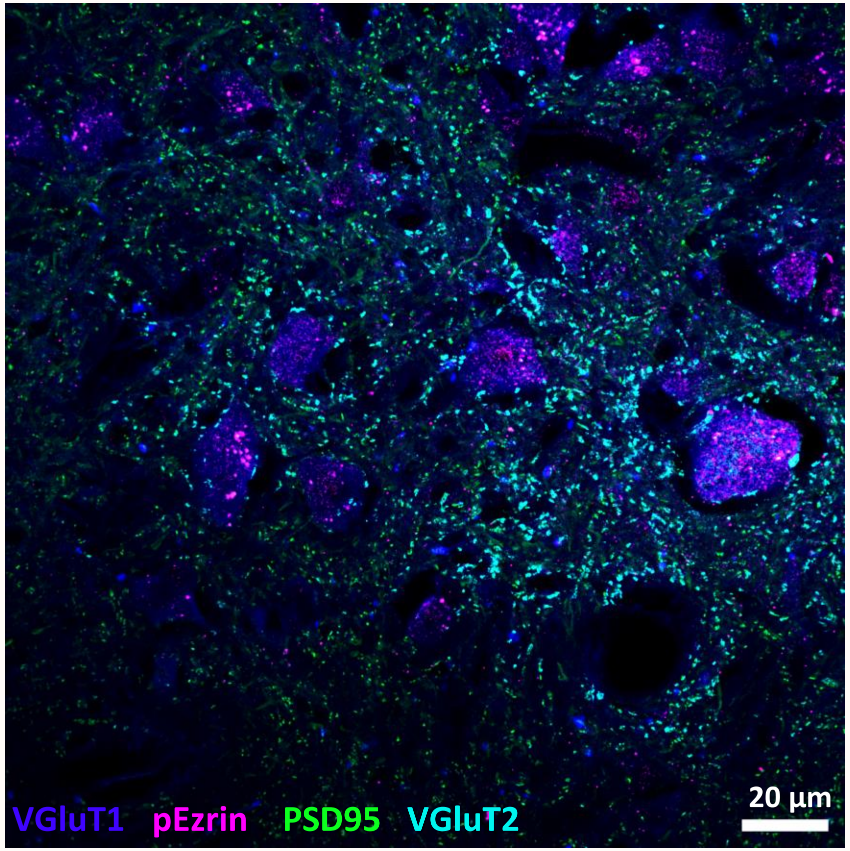


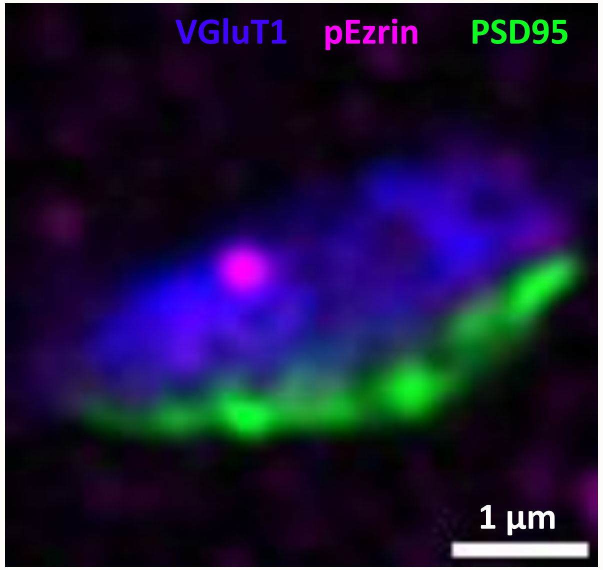


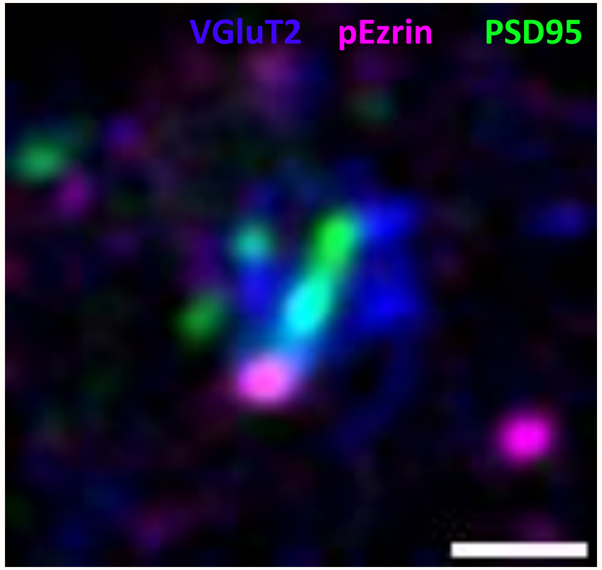
Imaging and Analysis

- Zeiss LSM800 Airyscan
  - High-resolution imaging
- FIJI (ImageJ)
  - Airyscan image processing

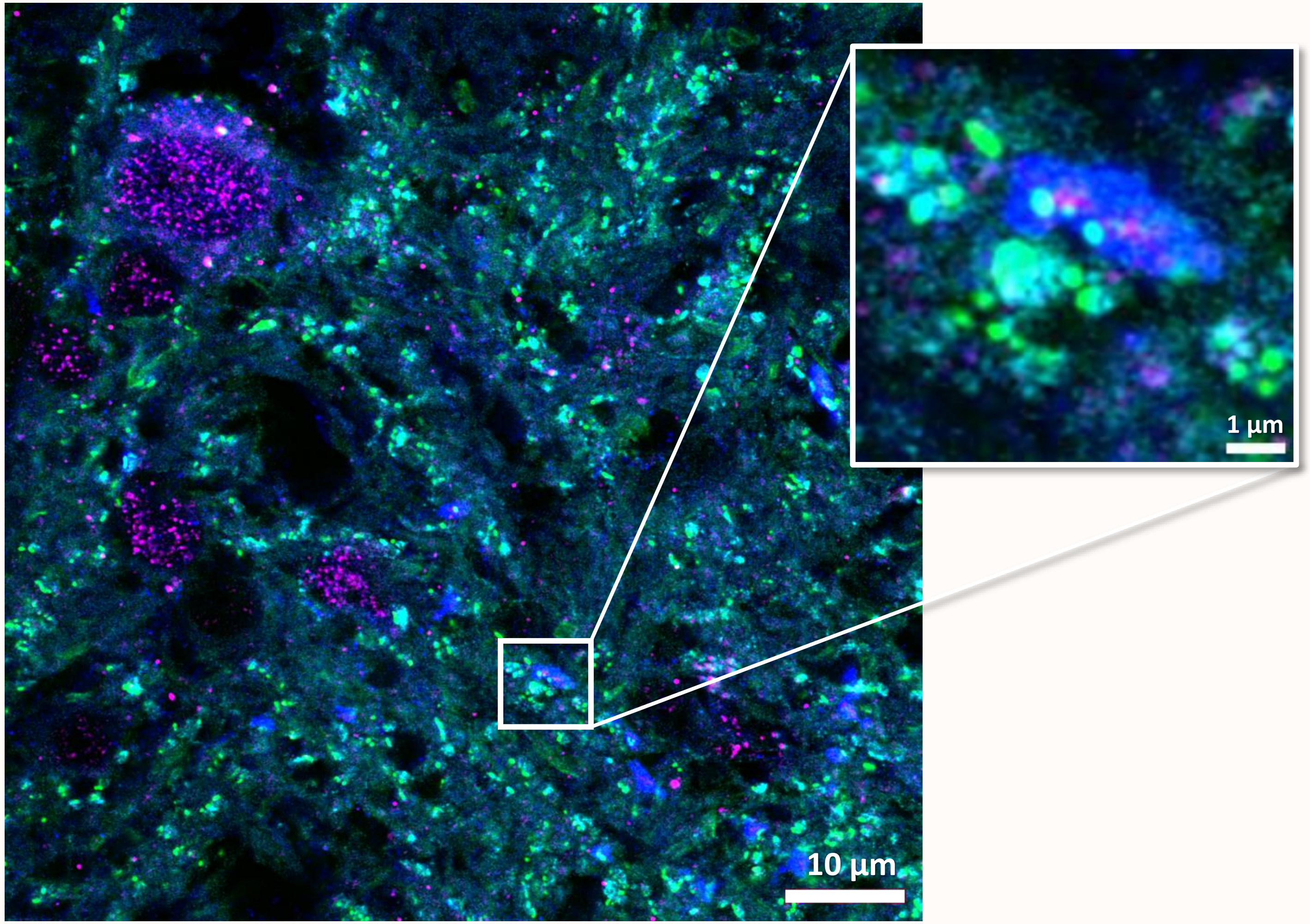




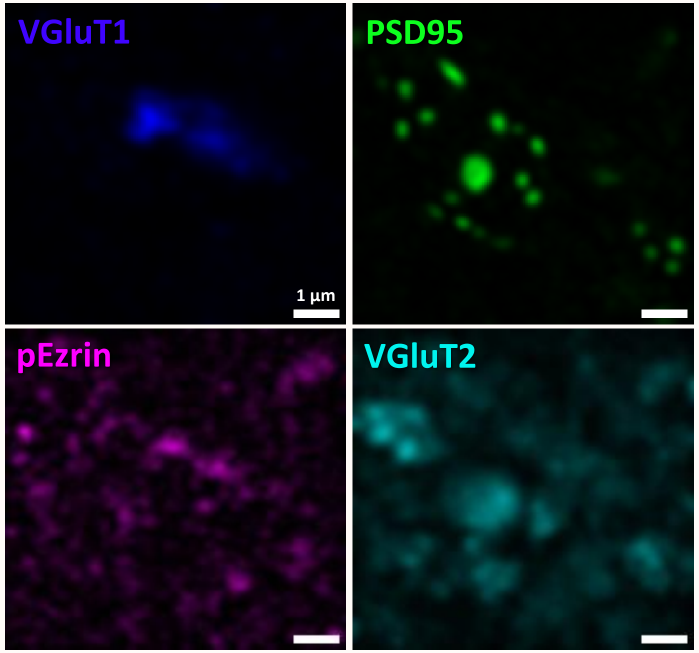




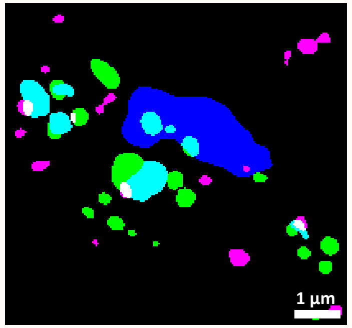
Imaging and Analysis

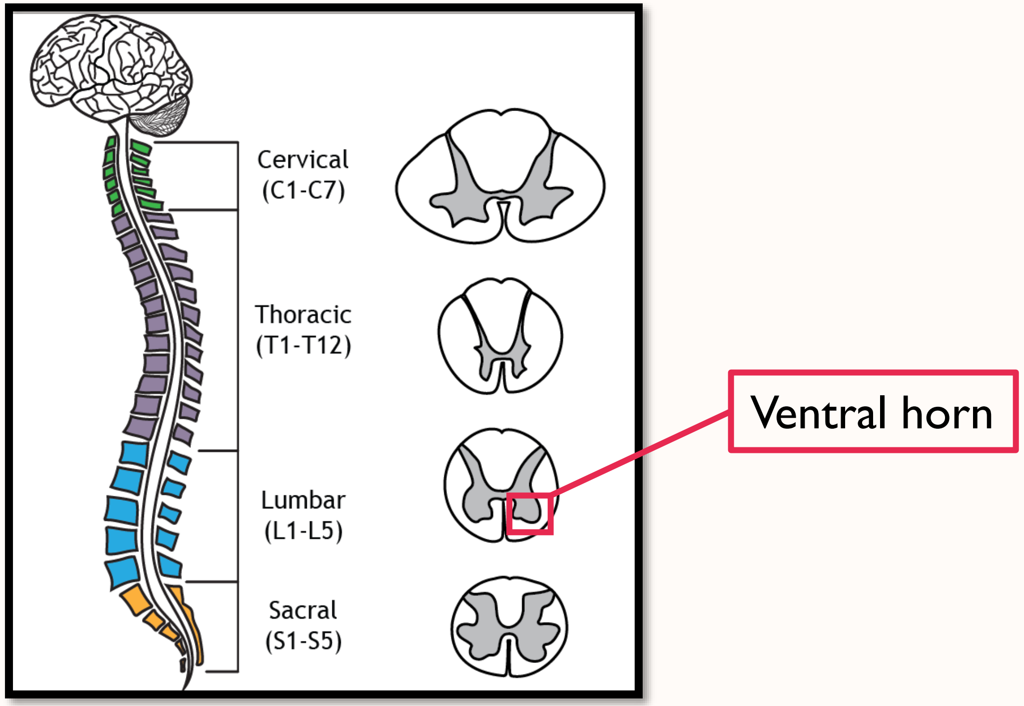


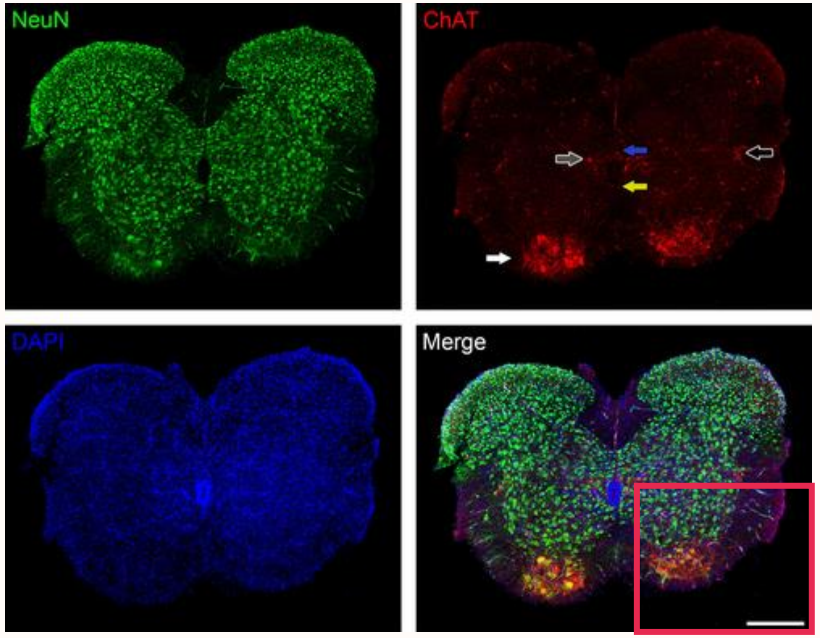
Background subtraction
Gaussian blurring

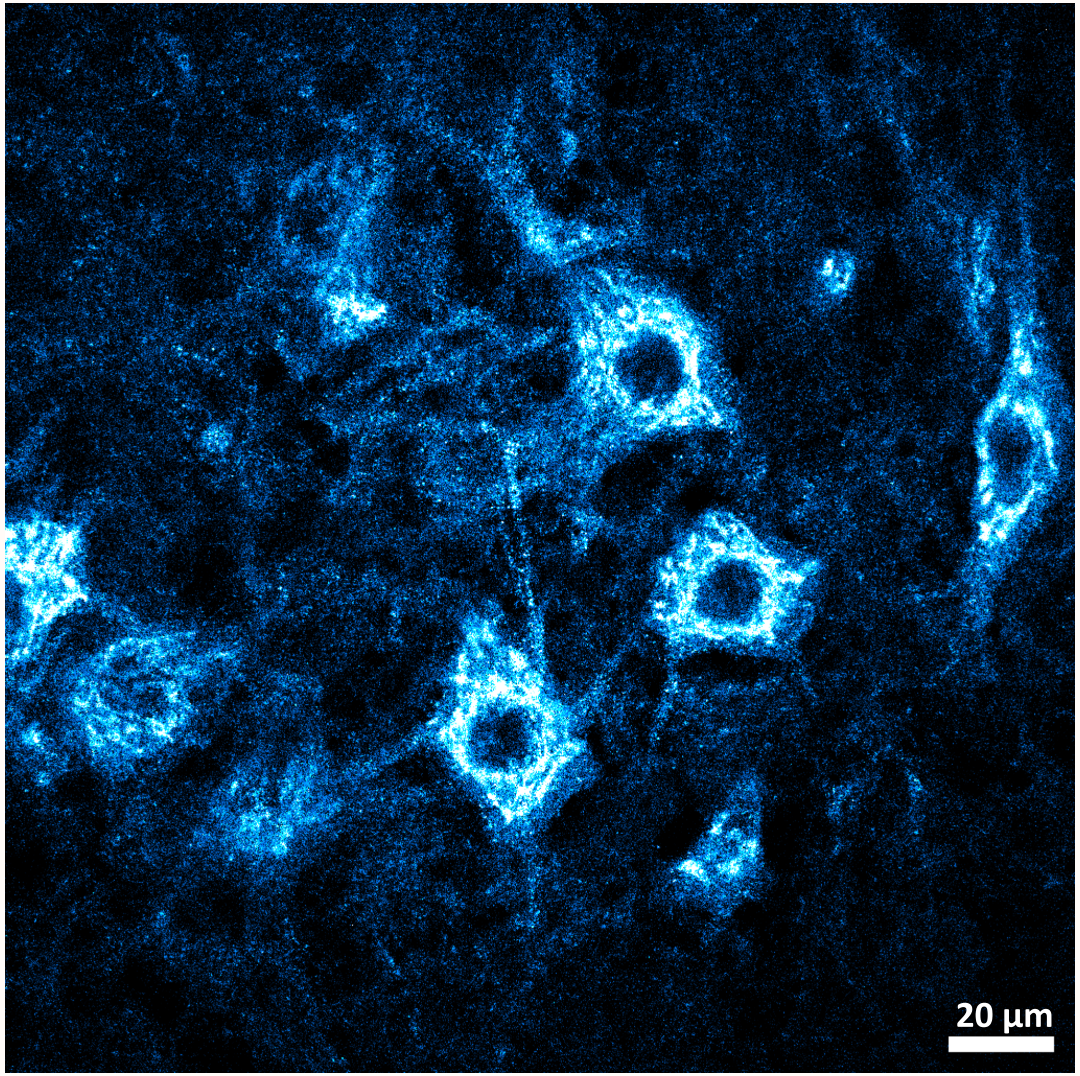


Thresholding
Binarisation



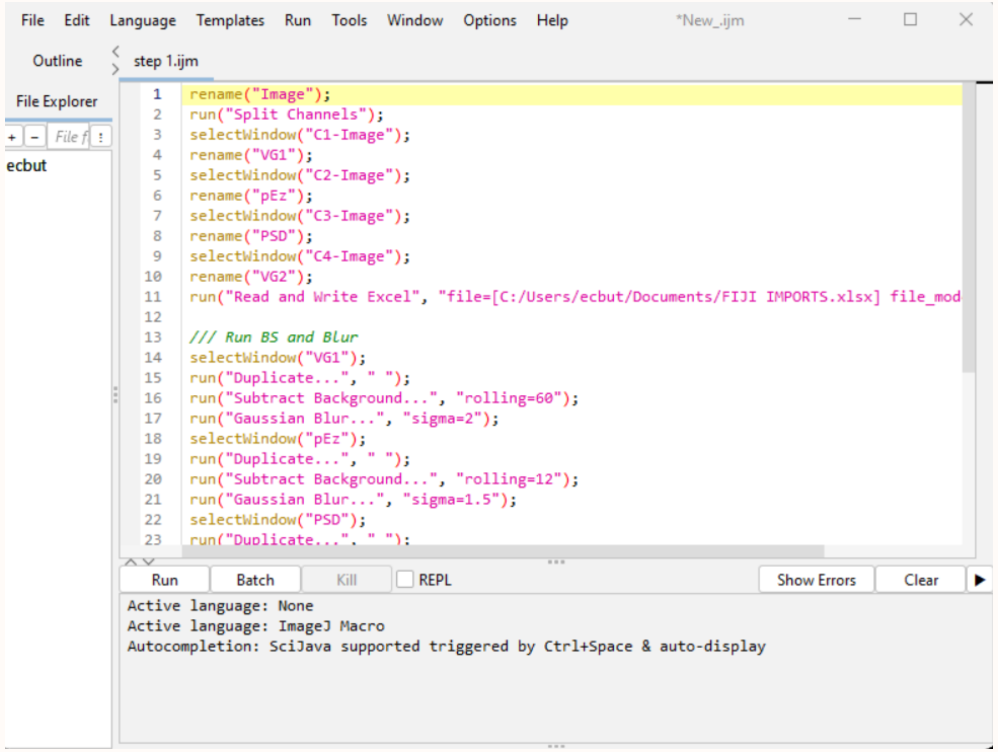

Cervical
(C1-C7)
Thoracic
(T1-T12)
Lumbar
(L1-L5)
Sacral
(S1-S5)
Ventral horn

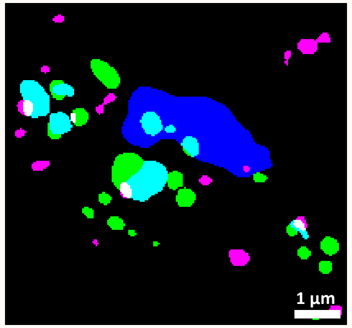

NeuN
ChAT
DAPI
Merge


20 µm

Imaging and Analysis

- Zeiss LSM800 Airyscan
  - High-resolution imaging
- FIJI (ImageJ)
  - Airyscan image processing
    - Size
    - Number
    - Colocalisation
    - Area
    - Density
    - Mean intensity
    - Circularity
    - Etc.





Imaging and Analysis

- Zeiss LSM800 Airyscan
  - High-resolution imaging
- FIJI (ImageJ)
  - Airyscan image processing
    - **Size**
    - **Number**
    - **Colocalisation**
- Python
  - Pandas package
  - Organise large datasets for analysis
- Excel and SPSS
  - Further statistical analysis

```
[1]: import numpy as np
import pandas as pd
import scipy as sp
import matplotlib as plt
import seaborn as sns
```

IMPORT DIRECTLY FROM IMAGEJ EXCEL FILE

```
[2]: _1A1 = pd.read_excel(io="Documents/TDP43 P2 Analysis.xlsx", sheet_name='Sheet1')
```

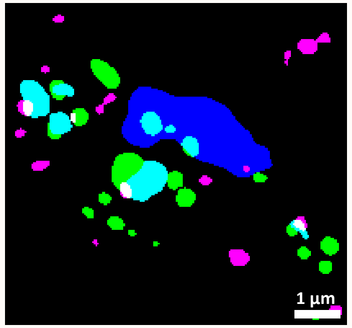
```
[3]: A11 = _1A1[['Unnamed: 1','Unnamed: 2','Unnamed: 13','Unnamed: 14','Unnamed: 25','Unnamed: 26']].drop(0).set_axis(['PSD AREA','PSD MEAN','SYN AREA','SYN M
A12 = _1A1[['Mask of PSD-1.2','Unnamed: 9','Unnamed: 10']].drop(0).set_axis(['PSD','PSD AREA','PSD MEAN'], axis='columns')
A13 = _1A1[['Mask of Syn-1.3','Mask of TDP-1.3']].drop(0).set_axis(['SYN','TDP'], axis='columns')
A14 = _1A1[['Mask of Syn-1.3', 'Unnamed: 37','Unnamed: 38','Mask of TDP-1.3', 'Unnamed: 41','Unnamed: 42']].drop(0).set_axis(['SYN','SYN AREA','SYN MEAN'
```

AREA SUM, COUNT, AND AVERAGE SIZE OF THREE MARKERS

```
[5]: A11['PSD AREA'] = pd.to_numeric(A11['PSD AREA'])
A11['SYN AREA'] = pd.to_numeric(A11['SYN AREA'])
A11['TDP AREA'] = pd.to_numeric(A11['TDP AREA'])
A12['PSD AREA'] = pd.to_numeric(A12['PSD AREA'])
A12['PSD MEAN'] = pd.to_numeric(A12['PSD MEAN'])
A14['SYN AREA'] = pd.to_numeric(A14['SYN AREA'])
A14['TDP AREA'] = pd.to_numeric(A14['TDP AREA'])


a1 = pd.DataFrame(A11[['PSD AREA','SYN AREA','TDP AREA']].sum(0)).set_axis(['Average Area'], axis='columns')
b1 = pd.DataFrame(A11[['PSD AREA','SYN AREA','TDP AREA']].count(0)).set_axis(['Average Count'], axis='columns')
c1 = pd.DataFrame(A11[['PSD AREA','SYN AREA','TDP AREA']].median(0)).set_axis(['Average Size'], axis='columns')
```

COUNTIF ANALYSIS OF TWO MARKERS WITHOUT USING LABEL COLUMNS



Future Directions

- Analysis of excitatory tripartite synapses is ongoing
- Investigate different types of tripartite synapses in TDP43ΔNLS model
  - Inhibitory
  - Cholinergic
- Importance: astrocytic changes due to ALS pathology have not been investigated in the TDP43ΔNLS model

Questions?

☺

REFERENCES

1. About ALS – ALS Society of Canada
2. Bar-Peled, O., O'brien, R. J., Morrison, J. H., & Rothstein, J. D. (1999). Clinical Neuroscience NeuroReport Cultured motor neurons possess calcium-permeable AMPA/kainate receptors. Clinical Neuroscience, 10(4), 855–859. http://journals.lww.com/neuroreport
3. Williamson, T. L., & Cleveland, D. W. (1999). Slowing of axonal transport is a very early event in the toxicity of ALS-linked SOD1 mutants to motor neurons. Nature, 2(1), 50–56.
4. Witts, E. C., Zagoraiou, L., & Miles, G. B. (2014). Anatomy and function of cholinergic C bouton inputs to motor neurons. In Journal of Anatomy (Vol. 224, Issue 1, pp. 52–60). Blackwell Publishing Ltd. https://doi.org/10.1111/joa.12063
5. Andrews, J. A., Jackson, C. E., Heiman-Patterson, T. D., Bettica, P., Brooks, B. R., & Pioro, E. P. (2020). Real-world evidence of riluzole effectiveness in treating amyotrophic lateral sclerosis. In Amyotrophic Lateral Sclerosis and Frontotemporal Degeneration (Vol. 21, Issues 7–8, pp. 509–518). Taylor and Francis Ltd. https://doi.org/10.1080/21678421.2020.1771734
6. Jaiswal, M. K. (2019). Riluzole and edaravone: A tale of two amyotrophic lateral sclerosis drugs. In Medicinal Research Reviews (Vol. 39, Issue 2, pp. 733–748). John Wiley and Sons Inc. https://doi.org/10.1002/med.21528
7. Rosen, D. R., Siddiquet, T., Pattersont, D., Figlewicz, D. A., Sapp, P., Hentatit, A., Donaldsont, D., Goto, J., O, J. P., Dengt, H.-X., Rahmanit, Z., Krizus, A., McKenna-Yasek, D., Cayabyabt, A., Gaston, S. M., Bergert, R., Tanzi, R. E., Halperin, J. J., Herzfeldtt, B., … Brown Jr, R. H. (1993). Mutations in Cu/Zn superoxide dismutase gene are associated with familial amyotrophic lateral sclerosis. Nature.
8. Williamson, T. L., & Cleveland, D. W. (1999). Slowing of axonal transport is a very early event in the toxicity of ALS-linked SOD1 mutants to motor neurons. Nature, 2(1), 50–56.
9. Genetics of amyotrophic lateral sclerosis: an update - PMC (nih.gov)
10. Aousji, O., Feldengut, S., Antonucci, S., Schön, M., Boeckers, T. M., Matschke, J., Mawrin, C., Ludolph, A. C., Del Tredici, K., Roselli, F., & Braak, H. (2023). Patterns of synaptic loss in human amyotrophic lateral sclerosis spinal cord: a clinicopathological study. Acta Neuropathologica Communications, 11(1). https://doi.org/10.1186/s40478-023-01616-8
11. FYI: Familial Amyotrophic Lateral Sclerosis (FALS) and Genetic Testing | The ALS Association
12. Bak, A. N., Djukic, S., Kadlecova, M., Braunstein, T. H., Jensen, D. B., & Meehan, C. F. (2023). Cytoplasmic TDP-43 accumulation drives changes in C-bouton number and size in a mouse model of sporadic Amyotrophic Lateral Sclerosis. Molecular and Cellular Neuroscience, 125. https://doi.org/10.1016/j.mcn.2023.103840
13. Walker, A. K., Spiller, K. J., Ge, G., Zheng, A., Xu, Y., Zhou, M., Tripathy, K., Kwong, L. K., Trojanowski, J. Q., & Lee, V. M. Y. (2015). Functional recovery in new mouse models of ALS/FTLD after clearance of pathological cytoplasmic TDP-43. Acta Neuropathologica, 130(5), 643–660. https://doi.org/10.1007/s00401-015-1460-x

REFERENCES

14. Mitra, J., Dharmalingam, P., Kodavati, M. M., Guerrero, E. N., Rao, K. S., & Hegde, M. L. (2024). Endogenous TDP-43 mislocalization in a novel knock-in mouse model reveals DNA repair impairment, inammation, and neuronal senescence. Research Square. https://doi.org/10.21203/rs.3.rs-3879966/v1

15. San Gil, R., Pascovici, D., Venturato, J., Brown-Wright, H., Mehta, P., Madrid San Martin, L., Wu, J., Luan, W., Chui, Y. K., Bademosi, A. T., Swaminathan, S., Naidoo, S., Berning, B. A., Wright, A. L., Keating, S. S., Curtis, M. A., Faull, R. L. M., Lee, J. D., Ngo, S. T., … Walker, A. K. (2024). A transient protein folding response targets aggregation in the early phase of TDP-43-mediated neurodegeneration. Nature Communications, 15(1), 1508. https://doi.org/10.1038/s41467-024-45646-9

16. Scherz, B., Rabl, R., Flunkert, S., Rohler, S., Neddens, J., Taub, N., Temmel, M., Panzenboeck, U., Niederkofler, V., Zimmermann, R., & Hutter-Paier, B. (2018). MTh1 driven expression of hTDP-43 results in typical ALS/FTLD neuropathological symptoms. PLoS ONE, 13(5). https://doi.org/10.1371/journal.pone.0197674

17. Frontiers | TDP-43 and ER Stress in Neurodegeneration: Friends or Foes? (frontiersin.org)

18. Lobsiger, C. S., & Cleveland, D. W. (2007). Glial cells as intrinsic components of non-cell autonomous neurodegenerative disease. In Nature Neuroscience (Vol. 10, Issue 11, pp. 1355–1360). https://doi.org/10.1038/nn1988

19. Rothstein, J. D. (2009). Current hypotheses for the underlying biology of amyotrophic lateral sclerosis. In Annals of Neurology (Vol. 65, Issue SUPPL. 1). John Wiley and Sons Inc. https://doi.org/10.1002/ana.21543

20. Konsolaki, E., Koropouli, E., Tsape, E., Pothakos, K., & Zagoraiou, L. (2020). Genetic Inactivation of Cholinergic C Bouton Output Improves Motor Performance but not Survival in a Mouse Model of Amyotrophic Lateral Sclerosis. Neuroscience, 450, 71–80. https://doi.org/10.1016/j.neuroscience.2020.07.047

21. Zholudeva, L. V., Abraira, V. E., Satkunendrarajah, K., McDevitt, T. C., Goulding, M. D., Magnuson, D. S. K., & Lane, M. A. (2021). Spinal interneurons as gatekeepers to neuroplasticity after injury or disease. Journal of Neuroscience, 41(5), 845–854. https://doi.org/10.1523/JNEUROSCI.1654-20.2020

22. Kaplan, A., Spiller, K. J., Towne, C., Kanning, K. C., Choe, G. T., Geber, A., Akay, T., Aebischer, P., & Henderson, C. E. (2014). Neuronal matrix Metalloproteinase-9 is a determinant of selective Neurodegeneration. Neuron, 81(2), 333–348. https://doi.org/10.1016/j.neuron.2013.12.009

23. Marchetto, M. C. N., Muotri, A. R., Mu, Y., Smith, A. M., Cezar, G. G., & Gage, F. H. (2008). Non-Cell-Autonomous Effect of Human SOD1G37R Astrocytes on Motor Neurons Derived from Human Embryonic Stem Cells. Cell Stem Cell, 3(6), 649–657. https://doi.org/10.1016/j.stem.2008.10.001

24. Clement, A., Nguyen, M., Roberts, E., Garcia, M., Boillee, S., Rule, M., McMahon, A., Doucette, W., Siwek, D., Ferrante, R., Brown Jr., R., Julien, J., Goldstein, L., & Cleveland, D. (2003). Wild-Type Nonneuronal Cells Extend Survival of SOD1 Mutant Motor Neurons in ALS Mice. Science, 302, 113–117. https://www.science.org

25. Na Pramatarova, A., Laganiè, J., Roussel, J., Ri Brisebois, K., & Rouleau, G. A. (2001). Neuron Specific Expression of Mutant Superoxide Dismutase 1 in Transgenic Mice Does Not Lead to Motor Impairment. Journal of Neuroscience.

REFERENCES

26. Broadhead, M. J., Bonthron, C., Waddington, J., Smith, W. V., Lopez, M. F., Burley, S., Valli, J., Zhu, F., Komiyama, N. H., Smith, C., Grant, S. G. N., & Miles, G. B. (2022). Selective vulnerability of tripartite synapses in amyotrophic lateral sclerosis. Acta Neuropathologica, 143(4), 471–486. https://doi.org/10.1007/s00401-022-02412-9

27. Broadhead, M. J., & Miles, G. B. (2021). A common role for astrocytes in rhythmic behaviours? In Progress in Neurobiology (Vol. 202). Elsevier Ltd. https://doi.org/10.1016/j.pneurobio.2021.102052

28. Igaz, L. M., Kwong, L. K., Lee, E. B., Chen-Plotkin, A., Swanson, E., Unger, T., Malunda, J., Xu, Y., Winton, M. J., Trojanowski, J. Q., & Lee, V. M. Y. (2011). Dysregulation of the ALS-associated gene TDP-43 leads to neuronal death and degeneration in mice. Journal of Clinical Investigation, 121(2), 726–738. https://doi.org/10.1172/JCI44867

29. Walker, A. K., Spiller, K. J., Ge, G., Zheng, A., Xu, Y., Zhou, M., Tripathy, K., Kwong, L. K., Trojanowski, J. Q., & Lee, V. M. Y. (2015). Functional recovery in new mouse models of ALS/FTLD after clearance of pathological cytoplasmic TDP-43. Acta Neuropathologica, 130(5), 643–660. https://doi.org/10.1007/s00401-015-1460-x

30. Kukreja, L., Shahidehpour, R., Kim, G., Keegan, J., Sadleir, K. R., Russell, T., Csernansky, J., Mesulam, M., Vassar, R. J., Wang, L., Dong, H., & Geula, C. (2018). Differential neurotoxicity related to tetracycline transactivator and TDP-43 expression in conditional TDP-43 mouse model of frontotemporal lobar degeneration. Journal of Neuroscience, 38(27), 6045–6062. https://doi.org/10.1523/JNEUROSCI.1836-17.2018

31. San Gil, R., Pascovici, D., Venturato, J., Brown-Wright, H., Mehta, P., Madrid San Martin, L., Wu, J., Luan, W., Chui, Y. K., Bademosi, A. T., Swaminathan, S., Naidoo, S., Berning, B. A., Wright, A. L., Keating, S. S., Curtis, M. A., Faull, R. L. M., Lee, J. D., Ngo, S. T., … Walker, A. K. (2024). A transient protein folding response targets aggregation in the early phase of TDP-43-mediated neurodegeneration. Nature Communications, 15(1), 1508. https://doi.org/10.1038/s41467-024-45646-9

32. Cytoplasmic TDP-43 accumulation drives changes in C-bouton number and size in a mouse model of sporadic Amyotrophic Lateral Sclerosis – ScienceDirect

33. The Postsynaptic Organization of Synapses - PMC (nih.gov)

34. Broadhead, M. J., Bonthron, C., Arcinas, L., Bez, S., Zhu, F., Goff, F., Nylk, J., Dholakia, K., Gunn-Moore, F., Grant, S. G. N., & Miles, G. B. (2020). Nanostructural Diversity of Synapses in the Mammalian Spinal Cord. Scientific Reports, 10(1). https://doi.org/10.1038/s41598-020-64874-9

35. Vesicular glutamate transporters are H+-anion exchangers that operate at variable stoichiometry | Nature Communications